

ALL INDIA DISCOMs ASSOCIATION (AIDA)

Minutes of 3rd Meeting of Committee on Specifications and Procurement Practices

Minutes of 3rd Meeting of Committee on Specifications and Procurement Practices held virtually on 09th October, 2025 at 12.00 noon.

Welcome and Introduction

The Director (Regulatory Affairs), AIDA, initiated the meeting and briefed the members on the background of the draft specification. He informed that the preparation began based on the draft Distribution Transformer Guidelines framed by the Central Electricity Authority (CEA). The draft was subsequently discussed in the first Committee meeting held on 17.06.2025, where Mr. Vivek Goel (CEA) provided an initial outline.

The draft specification was thereafter circulated to all DISCOMs for comments, and feedback received from TPDDL, BYPL, BRPL, MSEDCL, APEPDCL, IEEMA and PGVCL. These inputs were deliberated upon in the following committee meeting held on 04.09.2025. Based on the feedback and further suggestions, the AIDA Secretariat revised the draft, which was circulated to all member Discoms on 1st October, 2025.

Comments have since been received from CEA and BYPL, which have been structured for discussion in this Committee meeting (9th October, 2025), in which 42 nos. of representatives from various Discoms and IEEMA participated.

Opening Remarks by DG, AIDA

Shri Alok Kumr, DG, AIDA welcomed all participants and emphasized that the initiative aims to develop a harmonized standard specification to improve quality and reduce costs of distribution equipment. He informed that the Chairman, CEA, Shri Ghanshyam Prasad appreciated the effort and nominated Shri Vivek Goel, Chief Engineer (CEA) to support it.

During the meeting, the following points/comments were discussed with the respective Members of Discoms:

1. On Page 2 (Introduction) of the draft, the reference to ISO 9001:2015 and ISO-12000 for maintaining a safe working environment was discussed. CEA commented that ISO-12000 is not relevant in this context and recommended replacing it with ISO 45001:2018, suggesting that the text emphasize adherence to ISO 9001:2015 for quality management and ISO 45001:2018 for occupational health and safety. The suggestion made by CEA was agreed upon and it was decided that the clause shall be revised accordingly.
2. On Page 5 of the draft specification titled "*Standard Technical Specification for Single Phase Distribution Transformers 11/V3 kV/250V or 11kV/250V (Non-sealed, Outdoor Type)*", CEA commented that as per IS 1180, there is no provision for non-sealed single-phase distribution transformers and recommended replacing "non-sealed, outdoor type" with "sealed type, outdoor." BYPL also supported this view. During the meeting, DISCOM members agreed with CEA's recommendation and suggested that single-phase DTs should be of sealed type as per IS 1180.
3. On Page 9 of the draft under "Service Conditions – Humidity 95%, Average Rain 1500 mm", CEA commented that these parameters should be determined by respective States based on



local climatic conditions due to wide regional variations. During the meeting, DISCOM members suggested consulting manufacturers to assess any cost implications of designing for worst-case humidity and rainfall conditions.

4. On Page 11 of the draft under "Winding – 2.2.2", which states that only pre-approved makes of winding and insulation materials shall be used, CEA commented that such a requirement may not be necessary for distribution transformers, though it should be verified whether DISCOMs maintain pre-approved lists. During the meeting, DISCOM members suggested not to include a pre-approved list in the specification, and instead, define the material specifications clearly, ensuring that all materials are fully type tested as per relevant standards.
5. On Page 12 of the draft under "Oil – 2.3.3", which specifies that only pre-approved makes of insulating oil shall be used, CEA commented that such a clause may not be necessary for distribution transformers, though it may be relevant for power transformers, and suggested verifying whether DISCOMs maintain pre-approved lists. During the meeting, DISCOM members suggested not to include a pre-approved list in the specification and instead define detailed material specifications, ensuring that all materials, including insulating oil, are fully type tested as per relevant standards.
6. On Page 12 of the draft under "Losses – 3.2", CEA recommended that distribution transformers (DTs) of 100 kVA and above should have Energy Efficiency (EE) level 2, while smaller DTs below 100 kVA should have EE level 1. During the meeting, DISCOM members suggested adopting EE level 2 for DTs of 100 kVA and above, and EE level 1 for DTs below 100 kVA, considering cost implications.
7. On Page 12 of the draft under "Temperature Rise – 5.1", CEA recommended replacing the specified top oil and winding temperature rise limits with a clause referring to IS 1180. During the meeting, DISCOM members suggested adopting the temperature rise limits as per IS 1180.
8. On Page 13 of the draft under "Bushings – 7.4", CEA recommended that, as per IS 1180, the nuts should be made of stainless steel. During the meeting, DISCOM members suggested retaining brass nuts if the cost is not significantly higher, subject to confirmation with DT manufacturers.
9. On Page 13 of the draft under "Bushings – 7.5", the specification required high-quality gapless metal-oxide surge type lightning arrestors (LA) to be mounted on the transformer with separate grounding. CEA noted that this requirement is not mentioned in the Standard Bidding Document and requested justification. During the meeting, DISCOM members suggested not mounting the LA on the transformer due to potential damage to bushings, and recommended installing the LA separately on the pole.
10. On Page 13 of the draft under "Bushing Terminals – 8.1", the specification detailed provisions for HV and LV bushing connections, including heat-shrink covers, detachable connectors, and type-tested terminal connectors. CEA requested justification for any deviation from the Standard Bidding Document. During the meeting, DISCOM members suggested retaining provisions for HV conductor/cable connections and LV cable connections, allowing flexibility in design as per DISCOM requirements.
11. On Page 19 of the draft under "Tests – 17", the clause required bidders to conduct type tests at their own cost if the offered transformer was not type-tested. CEA suggested removing this, noting that type tests should be completed prior to bidding to avoid delays. AIDA agreed that all type tests should ideally be completed in advance.



12. On Page 20 of the draft under "*Routine Tests – 18.8*", CEA suggested that the separate source voltage (SSV) withstand test should not be applicable for single-phase DTs with 11/V3 kV primary voltage. During the meeting, DISCOM members suggested that all routine tests, including SSV, should be conducted as per IS standards.
13. On Page 42 of the draft under "*Standard Ratings – 3*", CEA recommended including 160 kVA, 315 kVA, and 630 kVA ratings and removing 800 kVA, as per IS 1180. During the meeting, DISCOM members suggested incorporating 630 kVA and deleting 800 kVA from the standard ratings.
14. On Page 46 of the draft under "*Tapping Ranges and Methods – 8.1.2*", CEA recommended that tapping arrangements follow IS 1180, with four taps for DTs below 200 kVA and seven taps for DTs above 200 kVA. During the meeting, DISCOM members suggested providing five taps for DTs above 100 kVA and seven taps for DTs above 500 kVA.
15. On Page 50 of the draft under "*Corrugated Tank – 17.3*", the specification allowed bidders to offer corrugated tanks for DTs up to 100 kVA. CEA noted that no reference was found for the 100 kVA limit and suggested seeking justification from DISCOMs. During the meeting, DISCOM members suggested retaining standard tanks with radiators for all DTs and removing the option of corrugated tanks from the specification.
16. In the draft specification, it was mentioned that winding shall be aluminum and standard ratings of single-phase transformers shall be 10, 16, and 25 kVA. BYPL suggested that the winding material could be aluminum or copper as per utility requirements, and noted that 50 kVA, 75 kVA, and 100 kVA are also mentioned in the latest amendment of IS 1180 (Part 1). During the meeting, DISCOM members suggested retaining aluminum winding for DTs up to 100 kVA and using copper winding for DTs above 100 kVA.
17. The materials under discussion included IS-2099 – Specification for high voltage porcelain bushing and IS-6600 – Guide for loading of oil-immersed transformers. BYPL noted that IS-2099 and IS-6600 have been superseded by IS/IEC 60137-2017 and IS 2026, Part 7, respectively, and AIDA acknowledged the comments. During the meeting, it was suggested that the references be updated and the Seismic Zone be incorporated.
18. The material specifies that the percentage impedance of single-phase transformers at 75 °C for ratings up to 25 kVA shall be as per Table 9 of IS 1180 (Part-1):2014. BYPL noted that percentage impedance for 50 kVA, 75 kVA, and 100 kVA is also included in the latest amendment of IS 1180 (Part-1). During the meeting, it was suggested to maintain the values as per IS 1180.
19. The transformer shall be suitable for loading as per the latest IS 6600, which has been superseded by IS 2026, Part 7. During the meeting, DISCOMs members suggested that the reference be corrected to IS 2026, Part 7.
20. In the draft, the color shade is specified as 541 of IS:5. There was a comment that the color shade shall be as per the requirement of the utility. During the meeting, DISCOMs members discussed this point, and DG, AIDA suggested that 2–3 choices of color shall be provided and DISCOMs can choose the color from the choices provided in the specifications.
21. There was a comment that an inbuilt protection for HV side & LV side shall be provided in the transformer. This was discussed in the meeting & some members shared that these inbuilt protections create maintenance related issues. Therefore, it was suggested that there shall be no inbuilt protection in the transformer. All the protections on HV & LV side shall be installed separately outside the distribution transformer.

22. The material under discussion is three-phase distribution transformers, 11 kV/433–250 V (non-sealed, outdoor type). As per IS 1180 Part 1, three-phase transformers can be either sealed or non-sealed based on utility requirements. During the meeting, DISCOMs members discussed this and advised that, for standardization and common usage, all three-phase distribution transformers shall be of the non-sealed type only.
23. The material under discussion is three-phase distribution transformers, 11 kV/433–250 V (non-sealed, outdoor type). The transformer rating can be 11 kV/415 V or 11 kV/433 V as per utility requirements. During the meeting, DISCOMs members noted that most DISCOMs use 433/250 V as the secondary voltage, particularly in suburban and rural areas, but 415/240 V may also be chosen without any cost impact. It was suggested to include both voltage options in the specifications, allowing DISCOMs to select the secondary voltage of 433 V or 415 V when issuing the purchase order.
24. The material under discussion is IS-2099 – Specification for high voltage porcelain bushing. BYPL noted that IS-2099 has been superseded by IS/IEC 60137. During the meeting, DISCOMs members suggested that the reference be corrected to IS/IEC 60137-2017.
25. It was proposed that the Seismic Zone be included in the specification. During the meeting, DISCOMs members noted this and suggested that it be incorporated.
26. For transformers above 1000 kVA, Buchholz relay is mandatory as per CEA safety guidelines. During the meeting, DISCOMs members suggested that the requirement should follow the relevant IS standards. Additionally, the transformer shall be suitable for loading as per the latest IS 6600, which has been superseded by IS 2026, Part 7. DISCOMs members suggested that the reference be corrected to IS 2026, Part 7.
27. For LT metering, it was suggested that LV side metering CTs shall be installed separately in the metering box or LTDB box. For marshalling boxes, DISCOMs members suggested that BZ, OTI, and WTI shall be provided for distribution transformers above 1000 kVA in a marshalling box, and similarly, OTI and WTI shall be included as per IS for these transformers. Regarding applicable standards, it was noted that the CBIP Manual on Transformers and CEA guidelines should be added to the list of applicable standards.

The draft shall be finalized after incorporating the inputs discussed during the meeting.



(Satyendra Nath Kalita)
Convenor, AIDA Committee